DMFR-YOLO: Defect Morphology Encoding and Feature Reassembly for Fabric Defect Detection

Kangzhao Tan[#], Jiarui Zhang[#], Junjie Yu, Heng Wu, Jing Lin[*], and Chunhua He[*]

***Abstract*—Circular knitting machines are widely used in continuous fabric production. Needle or yarn abnormalities can cause repeated defects along the fabric before they are detected, resulting in material waste and economic losses. To enable in-process defect detection and intervention, we develop a closed-loop defect detection system with edge inference and wired machine-stop control, and propose a lightweight detector termed DMFR-YOLO. DMFR-YOLO introduces defect morphology encoding (DME) into the backbone to explicitly model horizontally elongated defects, vertically elongated defects, and localized hole-like defects. In the feature pyramid, content-adaptive feature reassembly (CAFR) replaces fixed upsampling with content-adaptive reconstruction, improving the preservation of defect boundaries and weak responses during multi-scale feature fusion. Evaluated on a self-collected fabric defect dataset from a real-world circular knitting machine production line, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively, with 3.39 M parameters. The detector is deployed as an INT8 RKNN model on an RK3566 edge device. The resulting system integrates on-device detection, local stop control, operator review, and cloud-based quality analysis to support intervention during circular-knitting production.**

***Index Terms*—Industrial defect detection, YOLO, deep learning, edge computing, Internet of Things.**

I. Introduction

CIRCULAR knitting machines are important production infrastructure for high-volume weft-knit fabrics used in T-shirts and sportswear [1]–[3]. The International Textile Manufacturers Federation reported approximately 24,500 global shipments of large circular knitting machines in 2025 [4]. Globally, approximately 92 million tonnes of textile waste are generated annually, highlighting the importance of reducing avoidable material losses [5]. Conventional quality checks are often performed after knitting is completed. A persistent needle or yarn abnormality can therefore generate repeated defects along a fabric roll before discovery, accumulating substantial fabric waste and economic losses [6]–[8]. Real-time detection during knitting is needed to identify these defects and enable intervention before further defective fabric is produced.

Edge computing has been applied to distinct textile-inspection settings. Song et al. deploy EfficientDet-D0 on an NVIDIA Jetson TX2 and evaluate five fabric datasets: dark red, grid, camouflage, digitally printed, and light blue fabrics; the fabric-formation process is not specified [9]. Machado et al. deploy YOLO11n on a Jetson Orin Nano to inspect solid-color cotton and linen fabrics on a fabric rolling line [10]. Yan et al. deploy an improved YOLOv5s model on a Huawei Ascend 310-based Atlas platform for inspecting warp-knitted lace on warp-knitting machines [11]. These textile applications concern different materials and inspection settings. This work specifically addresses in-process inspection of circular weft-knitted fabric on circular knitting machines.

In other industrial inspection tasks, Wang et al. deploy an improved YOLOv5s model on a Jetson Xavier NX for rail surface defect detection [12]. Li et al. deploy LiteYOLO-ID on a Jetson TX2 NX for insulator defect detection [13], and Zhao et al. implement KAM-Net on a Xilinx ZCU102 FPGA for glass defect segmentation [14]. As summarized in Table I, these studies mainly employ general-purpose edge-computing platforms or FPGA development boards. Integrating these platforms into a circular knitting machine requires application-specific camera, sensing, and control interfaces. For installation across many machines, the cost of integrating these interfaces is also a practical consideration, although platform reference prices alone do not establish complete-system cost. This motivates a dedicated edge device that integrates image acquisition, defect detection, and machine-stop control for circular knitting production.

At the algorithm level, YOLO-based detectors provide efficient inference and explicit defect localization [15]. Recent studies have incorporated YOLOv3 into an optical detection system for LCD module defects [16], combined template assistance and attention with YOLOv7 for paper defects [17], and introduced feature prefusion and mask guidance into a Scaled-YOLOv4-style detector for camera-decoration surface defects [18]. Lightweight convolution and structural reparameterization have further supported insulator and rail defect detection on embedded platforms [12], [13]. These works demonstrate the practicality of the YOLO route and improve defect representation or inference efficiency. For circular-knitted fabrics, our focus is to retain this efficiency while explicitly representing the characteristic shapes of defects in repetitive backgrounds.

The defects considered in this work appear as horizontally elongated defects, vertically elongated defects, and localized holes. Standard convolutional feature extraction can learn these patterns implicitly, but it does not explicitly separate their spatial responses. In addition, the nearest-neighbor upsampling used in conventional feature pyramids reconstructs higher-resolution features through fixed replication rather than local content-dependent aggregation. Preserving thin defect boundaries and weak local responses under a limited computational budget therefore motivates morphology-specific

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feature encoding and adaptive reconstruction during multi-scale fusion.

Based on these hardware and algorithm considerations, we design and implement a dedicated defect detection system for circular knitting machines and propose a lightweight detector, DMFR-YOLO. The system integrates an industrial camera with LED illumination, a Hall sensor, a coaxial video link, and a self-developed RK3566 edge board. Machine rotation activates detection, and a detected defect triggers a local wired stop command and displays the result for operator review. The cloud receives defect and device records for monitoring and production-quality analysis without participating in the stop decision [19]. Within the detector, defect morphology encoding (DME) introduces horizontal, vertical, and local branches into the backbone, while content-adaptive feature reassembly (CAFR) replaces fixed top-down upsampling with locally predicted neighborhood aggregation. The hardware integration and compact detector jointly support detection and intervention during knitting.

The main contributions of our work are as follows.

(1) We design and implement a dedicated detection system for circular knitting machines, integrating image acquisition, camera transmission, RK3566 edge inference, and local machine-stop control on application-specific hardware, and demonstrate its deployment on a real production line.
(2) We introduce DME to encode the characteristic morphologies of horizontal defects, vertical defects, and holes through horizontal, vertical, and local branches, providing a lightweight morphology-aware representation for circular-knitted defect detection.
(3) We incorporate CAFR into the feature pyramid to replace fixed upsampling with locally predicted reassembly weights, enabling content-adaptive reconstruction during multi-scale fusion.

The remainder of the paper first presents the system overview and its four subsystems, then describes DMFR-YOLO, its two components, and the model deployment procedure. The experimental sections then define the data and evaluation protocol before presenting detector comparisons and component ablations.

II. Defect Detection and Closed-Loop Control System

A. System Overview

We design and implement a dedicated defect detection and closed-loop control system for circular knitting machines, as shown in Fig. 1. Our system integrates a power adapter, a global-shutter camera with LED illumination, a SerDes image-transmission link, a Hall sensor, a self-developed RK3566 edge board, and a cloud platform. These components are organized into four functional subsystems: power supply, image acquisition and transmission, network communication, and processor-based local computation and control. The power subsystem supplies the mainboard, which in turn powers and controls the LED illumination, and the image subsystem delivers fabric images to the processor over the SerDes link. The processor performs local defect detection and machine-stop control. The network subsystem uploads defect images and records to the cloud and delivers alert notifications to workers, while also supporting equipment monitoring and quality analysis.

Fig. 2 shows the physical implementation of our system. The camera assembly (A) combines the image sensor and LED illumination. We design a camera-side interface board (B) that integrates the serializer and coaxial connector, and a mainboard (C) built around the RK3566. The mainboard integrates memory and flash storage, the camera deserializer and coaxial interface, a 4G module with SIM-card and antenna interfaces, wired input/output interfaces, and power-management circuitry. The wired input/output interfaces connect the Hall sensor and machine controller, while the power circuitry receives the adapter output and provides the LED supply under mainboard current control. This board-level integration brings image reception, local processing, network communication, machine control, and illumination control into a dedicated edge device.

The complete closed-loop workflow begins when a worker starts the circular knitting machine. The Hall sensor detects machine rotation and sends a signal to the processor, which switches on the camera and LED illumination and enters the detection state. When a fabric defect is detected, the processor first sends a wired stop command to the machine controller to interrupt further defect propagation. The system then uploads the defect image and associated record to the cloud, which pushes an alert notification to inform the worker that intervention is required. The worker inspects the reported defect, resolves the production fault, and restarts the machine; the rotation signal then reactivates inspection. This sequence connects production, rotation sensing, image acquisition, defect detection, machine stopping, defect reporting, worker notification, and corrective intervention into a complete production–inspection–control loop. The following subsections describe the four subsystems that implement this workflow.

B. Power Supply Subsystem

To accommodate different industrial supply voltages across countries and regions, we design a power adapter with a 100–500 V AC input range and a 24 V DC output rated at 3 A. The adapter first supplies this output to the mainboard. The mainboard then supplies the LED illumination. By adjusting the current delivered to the LEDs, the mainboard controls the illumination brightness to suit the image-acquisition conditions. This arrangement integrates LED power delivery and brightness control into the mainboard.

C. Image Acquisition and Transmission Subsystem

Rapid fabric motion requires imaging that preserves defect geometry and fine spatial detail. We therefore select a high-resolution global-shutter camera, which exposes all pixels simultaneously to avoid motion-induced distortion from sequential row exposure [20]. The selected camera uses a SmartSens SC130GS global-shutter sensor at 1280×1024 resolution(SmartSens Technology Co., Ltd., Shanghai, China). At a camera-to-fabric distance of approximately 10 cm, the field

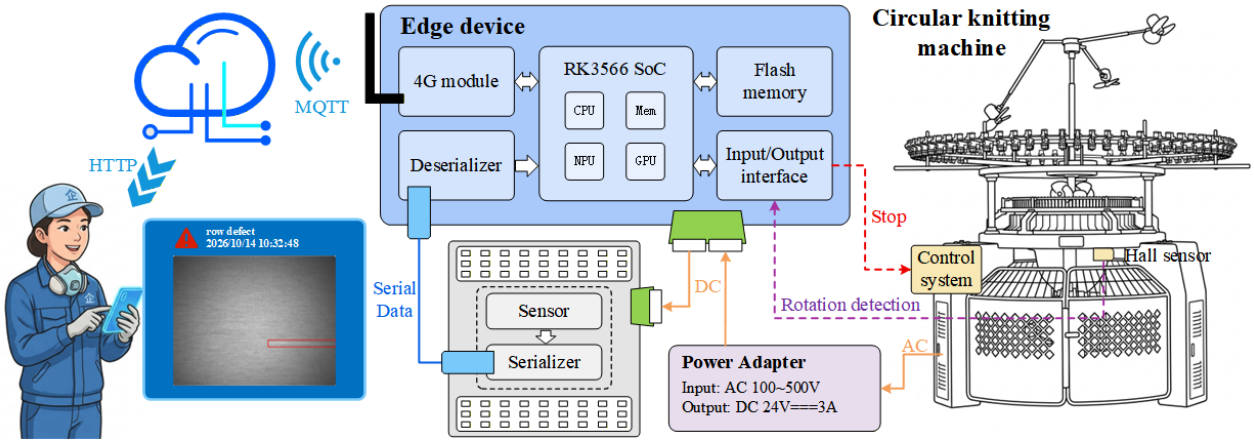


Fig. 1. System overview and subsystem interconnections. The power adapter supplies the mainboard, which powers the LED illumination and controls its brightness by adjusting the output current. Images are transmitted through the serializer, coaxial link, and deserializer to the RK3566 processor subsystem. The 4G network connects the device to the cloud for monitoring and defect alerts. Wired input/output interfaces receive the Hall-sensor rotation signal and send stop commands to the machine controller.

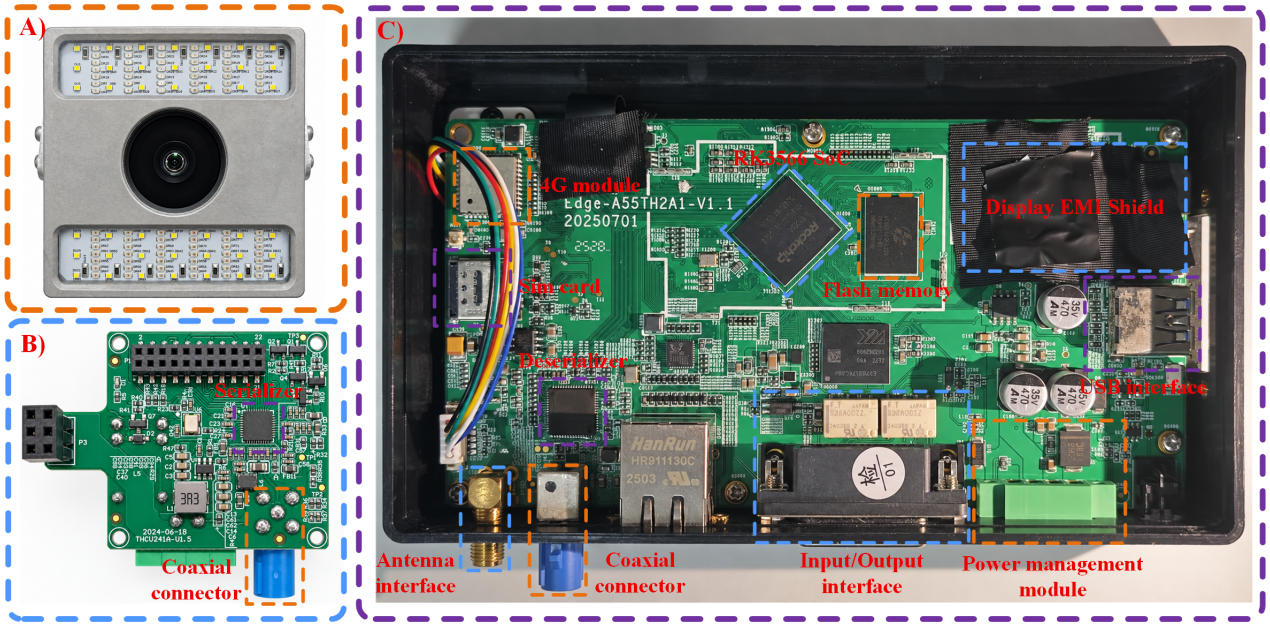


Fig. 2. Hardware of the production-line defect detection system: (A) industrial camera with LED illumination; (B) camera-side interface board with a serializer and coaxial connector; and (C) RK3566 edge board with the labeled communication, memory, camera-interface, input/output, and power-management components.

of view is approximately 16×12 cm, with the 16 cm dimension aligned with the horizontal direction of fabric motion.

Factory machinery creates an electromagnetically noisy environment, and installation requires the camera and edge device to be separated by a cable run. We therefore use serializer/deserializer (SerDes) transmission over shielded coaxial cable to accommodate remote camera placement and the interference environment [21]. Specifically, the THCV241A serializer and THCV242A deserializer bridge the camera-side and processor-side MIPI CSI-2 interfaces over a V-by-One HS link. The devices support a maximum serial line rate of 4.0 Gb/s per lane, corresponding to a rated effective rate of 3.2 Gb/s [22], [23].

The camera is configured to transmit 1280×1024 RAW10 images at 30 frames/s. With $b = 10$ bits per pixel, the active-image payload rate is

$$\begin{aligned} R_{\text{img}} &= WHbf_{\text{cam}} \\ &= 1280 \times 1024 \times 10 \times 30 \\ &= 393.216 \text{ Mb/s}, \end{aligned} \tag{1}$$

where W and H denote the image width and height, and f_{cam} is the camera acquisition rate. This corresponds to 49.152 MB/s, approximately 12.3% of the rated single-lane effective capacity, leaving nominal bandwidth headroom for transport overhead. The calculation excludes packet and framing overhead and compares the active-image load with the device rating. The 30 frames/s acquisition rate is distinct from the 10 frames/s minimum detection-throughput requirement derived below. The image stream is recovered at the edge device before being passed to the processor subsystem.

TABLE I
HARDWARE PLATFORMS USED IN RELATED DEFECT DETECTION STUDIES.

Platform	Memory (GB)	Compute capability	Price (USD)
Jetson TX2 [9]	8	1.33 TFLOPS (FP16)	599
Jetson TX2 NX [13]	4	1.33 TFLOPS (FP16)	249
Jetson Xavier NX [12]	8	21 TOPS (INT8)	399
Jetson Orin Nano [10]	8	40 TOPS (INT8)	499
Huawei Ascend 310 [11]	8	16 TOPS (INT8)	550
Xilinx ZCU102 [14]	4.5†	3.45 TOPS (INT8)‡	3234
Self-developed RK3566 edge device (Ours)	4	1 TOPS (INT8)	60

†ZCU102 memory is the sum of 4 GB PS DDR4 and 0.5 GB PL DDR4. ‡The 3.45 TOPS value is AMD's reference INT8 peak for three B4096 DPU cores.

D. Network Communication Subsystem

The network subsystem connects the edge device to the cloud platform over 4G. As illustrated in Fig. 1, MQTT carries device-state and defect-event messages between the device and cloud, while an HTTP interface gives operators access to monitoring information and defect records. The cloud supports device-state visualization, remote configuration, defect statistics, and production-quality analysis.

When the edge device detects a defect, it uploads the defect record and image to the cloud, which pushes a defect alert to the operator interface. The alert provides the defect category, occurrence time, and image for review and troubleshooting. These network functions complement the local display and wired stop command: defect inference and machine-stop decisions are executed by the processor subsystem without waiting for cloud communication.

E. Processor Subsystem

The processor is selected according to the inspection rate required by fabric motion and the camera field of view. For a design diameter of 40 in ($D = 1.016$ m) and a rotation period of $T = 2$ s, the fabric travels approximately $\pi D = 3.192$ m per revolution. With an effective field-of-view width of $w = 0.16$ m along the direction of motion, the required frame rate is

$$f_{\min} = \frac{\pi D}{Tw} = \frac{\pi \times 1.016}{2 \times 0.16} \approx 9.97 \text{ frames/s}. \tag{2}$$

We set the effective detection-throughput requirement to 10 frames/s, corresponding to 20 processed frames per revolution. Persistent needle or yarn abnormalities cause defects to propagate during knitting, providing further opportunities for detection in subsequent inspection cycles.

For this throughput target, we select the Rockchip RK3566 to balance processing capability, camera connectivity, and hardware cost. It integrates a quad-core Cortex-A55 CPU, an NPU rated at 1 TOPS for INT8 operations, and camera input interfaces [24]. Table I compares hardware specifications of the referenced platforms and our edge device. The 10 frames/s value is the system design requirement.

The detection algorithm and its deployment are introduced in the subsequent sections.

III. DETECTION ALGORITHM

A. DMFR-YOLO Architecture

DMFR-YOLO is developed based on YOLOv8n, which provides a compact, modular baseline and an existing Rockchip RKNN deployment reference [25], [26]. Within the local-computation setting established by the hardware platform, the proposed modifications focus on defect morphology encoding in the backbone and feature reconstruction in the feature pyramid.

Fig. 3 shows the overall architecture. DME is introduced after the main backbone stages to enhance the representation of different defect morphologies. The extracted multi-scale features are then passed to the neck for feature fusion [27], [28]. Within the feature pyramid, CAFR replaces two nearest-neighbor upsampling operations with content-adaptive feature reassembly to better preserve defect boundaries and weak responses. The fused features are subsequently used by the multi-scale detection head to predict defect locations and categories.

B. Defect Morphology Encoding

The three defect categories considered in this work present different spatial characteristics. Horizontal and vertical defects usually extend along one dominant direction, whereas hole defects are more localized. To capture these differences, we introduce a defect morphology encoding (DME) module with three parallel branches, as shown in the enlarged view in Fig. 3. The horizontal and vertical branches use 1×7 and 7×1 depthwise convolutions to enlarge the receptive field along the corresponding spatial directions, while the 3×3 branch captures localized defect structures. Depthwise convolution is adopted to model these spatial patterns with low computational overhead before channel-wise feature fusion.

For a C2f output $X \in \mathbb{R}^{C \times H \times W}$, the three branch responses are

$$F_h = \mathrm{DW}_{1\times7}(X), \tag{3}$$

$$F_v = \mathrm{DW}_{7\times1}(X), \tag{4}$$

$$F_l = \mathrm{DW}_{3\times3}(X). \tag{5}$$

Here, DW denotes depthwise convolution followed by batch normalization and SiLU activation. All branches preserve the spatial dimensions and produce C output channels. Their concatenation is fused as

$$F_d = \phi\left([F_h, F_v, F_l]\right), \tag{6}$$

where $[\cdot]$ denotes channel concatenation and ϕ is a 1×1 convolution mapping $3C$ channels to C, followed by batch normalization and SiLU. This learnable fusion combines the three types of spatial evidence rather than assigning a fixed

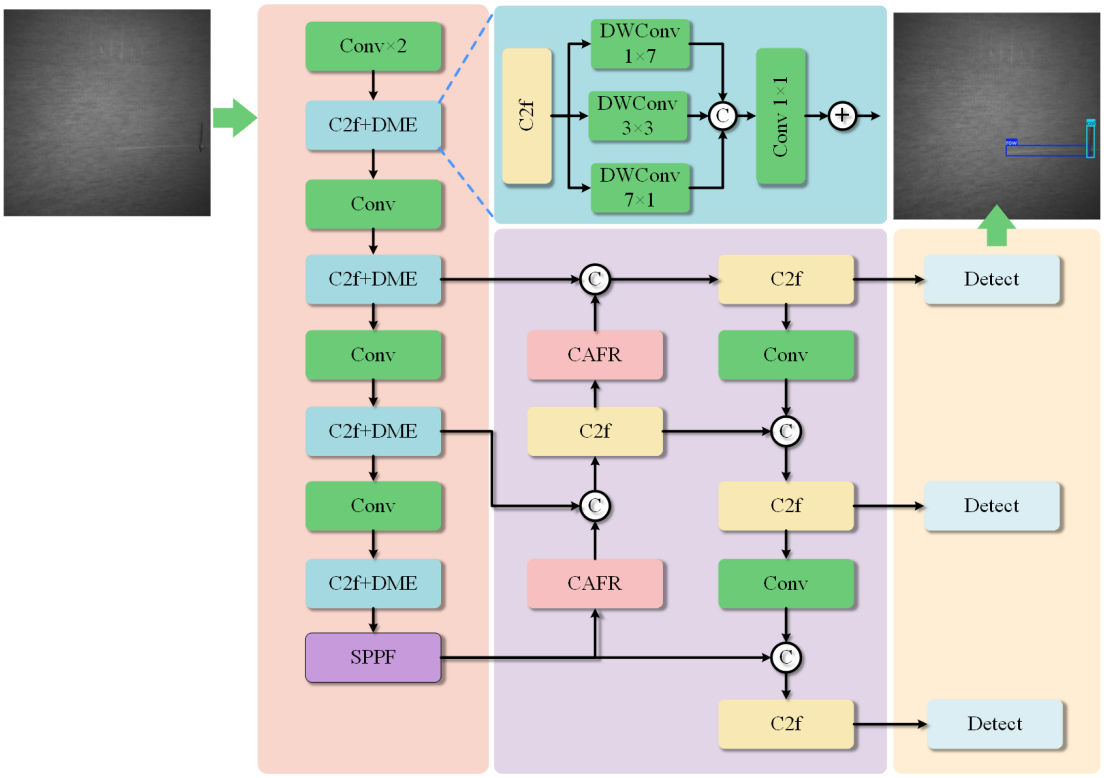


Fig. 3. Architecture of DMFR-YOLO with an enlarged view of DME. DME follows the backbone C2f stages, and CAFR replaces the two top-down upsampling operations. The neck fusion blocks, bottom-up path, and three-scale detection head are retained from YOLOv8n.

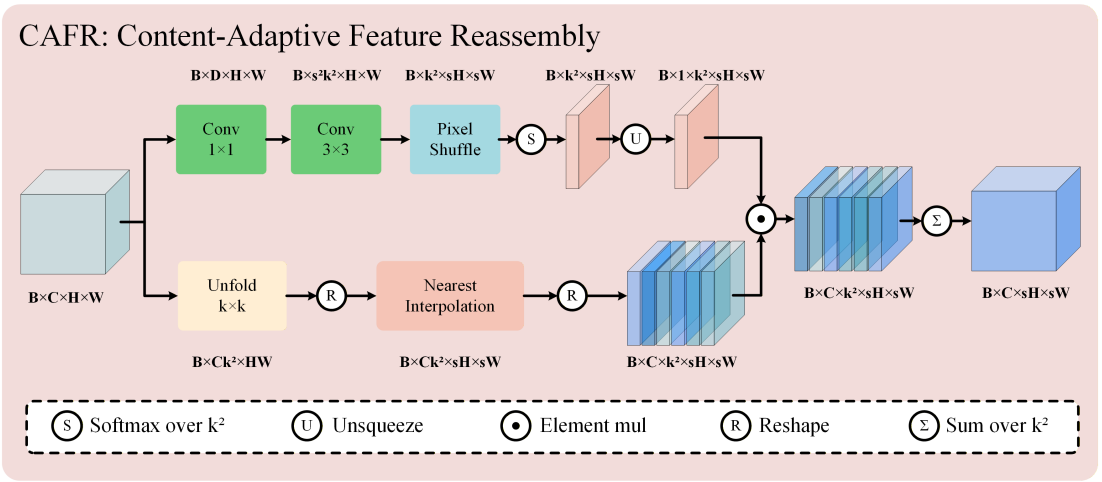


Fig. 4. CAFR structure. The upper path predicts location-specific 5×5 reassembly kernels, and the lower path unfolds local feature patches and replicates them to align with output locations. Weighted aggregation, rather than this alignment operation, produces the final $2\times$ upsampled feature.

branch to a particular defect class. A residual connection [29] gives

$$Y = X + F_d. \tag{7}$$

The residual connection combines the morphology-aware features with the incoming C2f output, allowing DME to enhance the backbone representation without discarding the features learned by the preceding layers. The directional branches are based on asymmetric convolution designs [30].

C. Content-Adaptive Feature Reassembly

In the feature pyramid, nearest-neighbor upsampling enlarges feature maps by directly replicating feature values. This fixed operation does not adapt to the spatial structure of defects. To address this limitation, we introduce content-adaptive feature reassembly (CAFR), following the reassembly mechanism of Content-Aware ReAssembly of FEatures (CARAFE) [31]. CAFR predicts spatially varying kernels from the input features and uses them to aggregate local neighborhoods (Fig. 4), aiming to better preserve defect boundaries and weak responses during multi-scale fusion.

In DMFR-YOLO, CAFR replaces the upsampling operations at both top-down fusion stages, with an upsampling factor of $s = 2$ and a reassembly kernel size of $k = 5$. As shown in Fig. 4, CAFR consists of two parallel branches for kernel prediction and local feature extraction. Their outputs are subsequently combined through weighted aggregation to generate the upsampled features. The two branches and their subsequent feature reassembly are described as follows:

1) Reassembly Kernel Prediction: Given an input feature map $X \in \mathbb{R}^{C\times H\times W}$, a 1×1 convolution first reduces its channel dimension:

$$X_c = E_{1\times1}(X). \tag{8}$$

Here, $X_c \in \mathbb{R}^{D\times H\times W}$, where $D = \max(\lfloor C/4 \rfloor, 16)$ denotes the compressed channel count; $E_{1\times1}$ includes batch normalization and SiLU activation. A linear 3×3 convolution then predicts the unnormalized reassembly weights:

$$K' = E_{3\times3}(X_c). \tag{9}$$

The resulting tensor is $K' \in \mathbb{R}^{s^2k^2\times H\times W}$. For $s = 2$ and $k = 5$, this corresponds to 100 weight channels at each input location. Pixel Shuffle rearranges the predicted weights onto the higher-resolution grid, followed by Softmax normalization:

$$K = \mathrm{Softmax}(\mathrm{PS}(K')). \tag{10}$$

Softmax is applied along the k^2 neighborhood dimension. The resulting tensor $K \in \mathbb{R}^{k^2\times sH\times sW}$ provides a normalized reassembly kernel for each output location.

2) Local Feature Extraction: In parallel, the patch branch extracts local neighborhoods from the input feature map using Unfold. The extracted patches are reshaped and upsampled using nearest-neighbor interpolation to match the spatial resolution of the predicted reassembly kernels:

$$\begin{aligned} P' &= \mathrm{Reshape}(\mathrm{Unfold}_{k\times k}(X)), \\ P &= \mathrm{Reshape}(\mathrm{Nearest}_s(P')). \end{aligned} \tag{11}$$

Here, $\mathrm{Unfold}_{k\times k}$ extracts a $k \times k$ neighborhood around each input position, and $\mathrm{Nearest}_s$ denotes nearest-neighbor upsampling with scale factor $s = 2$. The first reshape produces $P' \in \mathbb{R}^{Ck^2\times H\times W}$; the final reshape gives the aligned patch tensor $P \in \mathbb{R}^{C\times k^2\times sH\times sW}$, where k^2 indexes the positions within each local neighborhood. Nearest-neighbor interpolation is used only for neighborhood alignment, not to generate the final output features.

3) Feature Reassembly: The predicted kernels and the aligned local features are combined through element-wise multiplication and summation along the neighborhood dimension:

$$Y = \sum_{u=1}^{k^2} K_u \odot P_u. \tag{12}$$

Here, $K_u \in \mathbb{R}^{sH\times sW}$ and $P_u \in \mathbb{R}^{C\times sH\times sW}$ denote the weight map and corresponding local feature map for the u-th neighborhood position, respectively. The symbol $\odot$ denotes element-wise multiplication, with the weights broadcast across feature channels, producing an output feature map $Y \in \mathbb{R}^{C\times sH\times sW}$.

By predicting location-specific reassembly weights, CAFR adaptively aggregates local features during upsampling, helping preserve defect-related information in multi-scale feature fusion.

D. Model Deployment

DMFR-YOLO is deployed locally on the RK3566 edge device. After training, its PyTorch model is exported to ONNX with a fixed 640×640 input, then converted to an RKNN model using RKNN-Toolkit2 [32], with post-training INT8 quantization for efficient NPU execution [33]. An Android 13 application executes the RKNN model on the RK3566 edge device.

Fig. 5 shows the deployed system on the circular knitting machine production line. The on-device predictions feed the local display and wired stop-control workflow described in Section II-A.

IV. Experimental Setup

A. Dataset

Images were collected during operation of a real circular-knitting production line. A global-shutter industrial camera captures 1280×1024 frames approximately 10 cm from the fabric, with a field of view of approximately 16×12 cm, where the 16 cm dimension follows the circumferential motion. We selected images with visible defects and manually annotated their bounding boxes. The dataset includes three defect categories: horizontal defects (*row*), vertical defects (*col*), and holes (*hole*).

The dataset contains 403 images with 468 annotated defect instances, divided into training, validation, and test sets of 282, 40, and 81 images, respectively, as summarized in Table II. The validation set is used for checkpoint selection, while performance is reported on the test set containing 91 labeled defects. Because the selected images contain visible defects, this test set evaluates defect localization and classification; it does not measure false-stop frequency during continuous inspection of normal fabric.

TABLE II
IMAGE-LEVEL SPLIT AND DEFECT-INSTANCE COUNTS IN THE REAL PRODUCTION-LINE DATASET.

Split	Images	Instances	Row	Col	Hole
Train	282	329	174	75	80
Validation	40	48	24	13	11
Test	81	91	60	12	19

B. Implementation Details

Table III summarizes the training and deployment settings. DMFR-YOLO and YOLOv8n are initialized using the pre-trained yolov8n.pt weights, while newly added layers use default initialization. Other comparison models use the available pretrained checkpoints in their respective implementations. The remaining Ultralytics training settings follow the framework defaults.

The comparisons include lightweight YOLO models [25], [34], DETR [35], Deformable DETR [36], Faster R-CNN

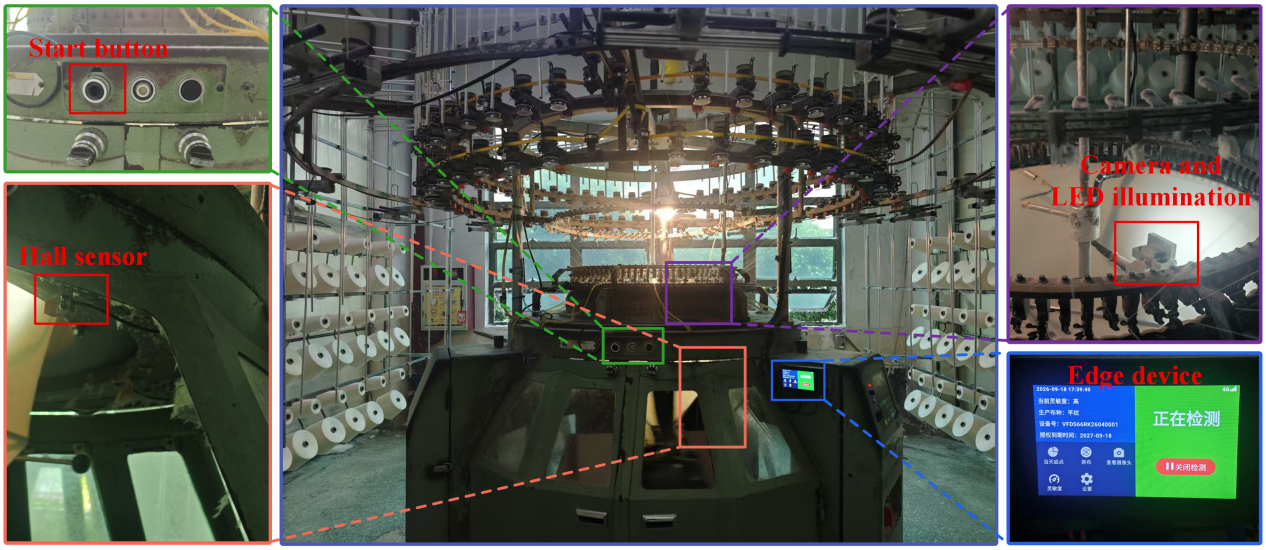


Fig. 5. Actual deployment on the circular knitting machine. The insets highlight the start button, Hall sensor, industrial camera, and RK3566 edge device, respectively.

TABLE III
IMPLEMENTATION PROTOCOL. TRAINING SETTINGS DESCRIBE THE ULTRALYTICS EXPERIMENTS; DEPLOYMENT SETTINGS DESCRIBE THE INSTALLED EDGE DEVICE.

Category	Parameter	Specification
Training setup	Hardware	Intel i7-13700KF; NVIDIA RTX 3090 (24 GB)
	Software	Ubuntu 24.04, Python 3.10.20, PyTorch 2.4.1+cu124, and Ultralytics 8.4.140
Deployment setup	Hardware	Rockchip RK3566
	System	Android 13
	Quantization	INT8
Data protocol	Input resolution	Fixed at 640×640
Training dynamics	Batch	16
	Maximum epochs	300
	Early-stopping patience	80 epochs

[37], and an R18 model from the RT-DETRv2 implementation, denoted RT-DETR-R18. All models are evaluated on the same test set using their respective frameworks and evaluation settings. The comparison therefore describes the evaluated model configurations rather than a controlled comparison under identical training and implementation conditions. The AP values in Section V characterize the test-set detector evaluation; they are not a separate accuracy measurement of the quantized RKNN model.

C. Evaluation Metrics

We report COCO average precision (AP) across ten IoU thresholds and AP at IoU 0.50 (AP50) [38]. For a predicted box B_p and a ground-truth box B_g, intersection over union is $\mathrm{IoU}(B_p, B_g) = |B_p \cap B_g|/|B_p \cup B_g|$. Let $\mathrm{AP}_c(t)$ denote the average precision for class c at IoU threshold t, and let N_c denote the number of defect categories. The metrics are defined as follows:

$$\mathrm{AP} = \frac{1}{10N_c} \sum_{c=1}^{N_c} \sum_{t \in T} \mathrm{AP}_c(t), \tag{13}$$

$$\mathrm{AP}_{50} = \frac{1}{N_c} \sum_{c=1}^{N_c} \mathrm{AP}_c(0.50), \tag{14}$$

where $T = \{0.50, 0.55, \ldots, 0.95\}$ and $N_c = 3$ in this study. All AP values are reported as percentages. AP therefore evaluates localization over increasingly strict overlap requirements, whereas AP50 uses a single, less restrictive threshold. Differences between reported AP values are expressed in percentage points.

Model parameters and FLOPs are measured using the respective frameworks with an input resolution of 640×640. These quantities describe model size and arithmetic complexity; on-device latency and machine stopping time require separate measurements.

TABLE IV
DETECTION RESULTS ON THE TEST SET. BOLD AND UNDERLINED VALUES INDICATE THE HIGHEST AND SECOND-HIGHEST REPORTED SCORES, RESPECTIVELY.

Model	Params (M)	FLOPs (G)	AP50 (%)	AP (%)
YOLOv5n	2.50	7.10	93.1	58.4
YOLO11n	2.58	6.40	95.0	59.6
YOLO12n	2.56	7.30	94.3	60.9
YOLO26n	2.38	5.30	96.5	61.4
RT-DETR-R18	20.09	60.72	95.2	**67.7**
DETR-R50	41.28	77.25	96.4	53.6
Deformable DETR-R50	39.82	158.66	<u>96.8</u>	64.3
Faster R-CNN-R50	41.13	181.45	96.1	64.7
YOLOv8n (Baseline)	3.01	8.10	94.1	59.3
DMFR-YOLO	3.39	8.90	**96.9**	<u>66.3</u>

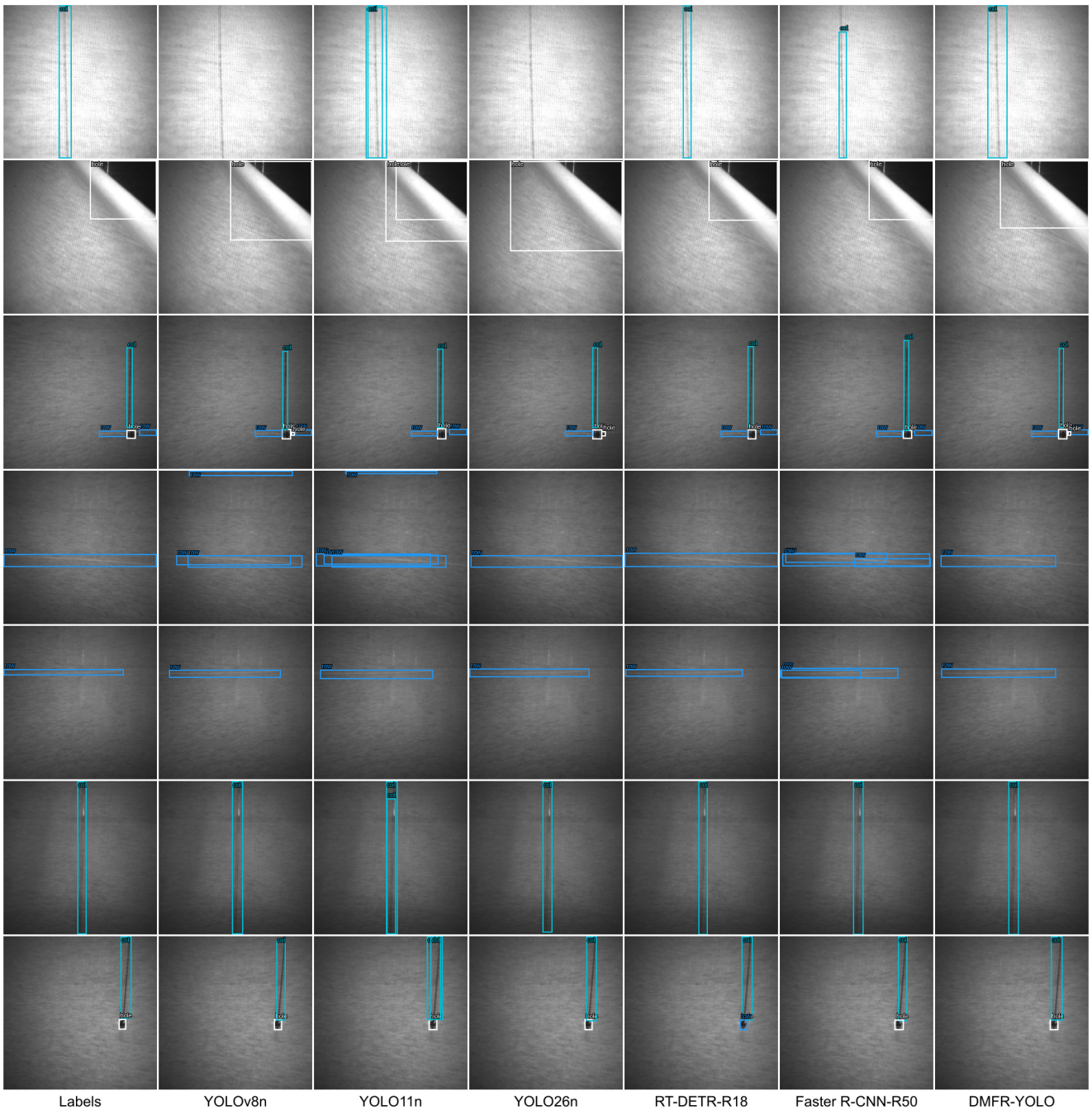


Fig. 6. Qualitative comparison of detection results on selected test images. From left to right, the columns show ground-truth annotations, YOLOv8n, YOLO11n, YOLO26n, RT-DETR-R18, Faster R-CNN-R50, and DMFR-YOLO (Ours).

V. Results and Discussion

A. Comparison With Other Detectors

Table IV presents the detection results on the test set. DMFR-YOLO achieves 66.3% AP and 96.9% AP50, improving the YOLOv8n baseline by 7.0 and 2.8 percentage points, respectively. These improvements are achieved with only 0.38 M additional parameters and approximately 0.8 G additional FLOPs. The AP gain reflects better aggregate performance across the evaluated IoU thresholds; the aggregate scores alone do not identify the source of the gain.

Among the evaluated lightweight YOLO detectors, DMFR-YOLO achieves the highest AP. Although RT-DETR-R18 attains a slightly higher AP of 67.7%, it requires 20.09 M parameters and 60.72 G FLOPs, compared with 3.39 M parameters and 8.90 G FLOPs for DMFR-YOLO.

Fig. 6 compares the detection results of different models on seven selected test images, highlighting differences in detection accuracy and bounding-box localization.

B. Component Ablation

Table V evaluates the individual and combined effects of DME and CAFR on detection performance. Introducing DME increases AP from 59.3% to 64.1%, while CAFR alone

TABLE V
COMPONENT ABLATION. PARAMETERS ARE IN MILLIONS, FLOPS IN BILLIONS, AND AP50/AP IN PERCENT.

Variant	DME	CAFR	Params	FLOPs	AP50	AP
YOLOv8n	–	–	3.01	8.1	94.1	59.3
+ DME	✓	–	3.28	8.8	94.2	64.1
+ CAFR	–	✓	3.11	8.2	94.9	60.7
DMFR-YOLO	✓	✓	3.39	8.9	**96.9**	**66.3**

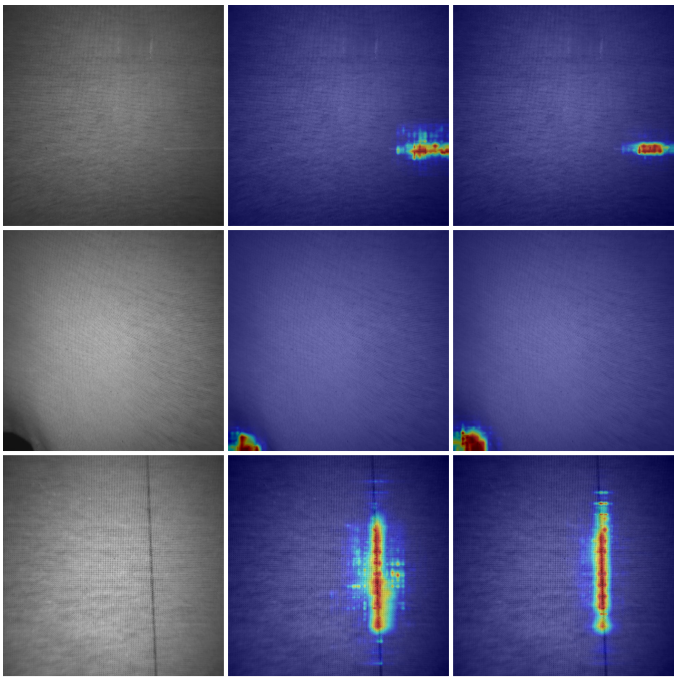


Fig. 7. Feature visualization of YOLOv8n and DMFR-YOLO. From left to right: original image (Origin), YOLOv8n, and DMFR-YOLO.

improves AP to 60.7%. When both modules are incorporated, DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming either individual modification. Within these evaluated configurations, the combined modification performs better than either module alone. This is consistent with the two modules acting at different stages: DME encodes spatial morphology in the backbone, while CAFR reconstructs features during multi-scale fusion. The aggregate ablation scores do not isolate the contribution of each directional branch or establish statistical significance across repeated training runs.

Fig. 7 compares the feature visualizations of YOLOv8n and DMFR-YOLO. Compared with YOLOv8n, DMFR-YOLO exhibits more concentrated feature responses around elongated defects and holes, with activation regions more closely aligned with the defect shapes. These observations suggest that DMFR-YOLO provides a more focused representation of defect regions.

VI. CONCLUSION

This paper presents DMFR-YOLO and a defect detection system for circular knitting machines. DMFR-YOLO introduces defect morphology encoding (DME) to capture the spatial characteristics of horizontal defects, vertical defects, and holes. Content-adaptive feature reassembly (CAFR) improves feature reconstruction during multi-scale fusion. Experiments on the production line dataset show that DMFR-YOLO achieves 66.3% AP and 96.9% AP50, outperforming YOLOv8n by 7.0 and 2.8 percentage points, respectively, with 3.39 M parameters and 8.9 G FLOPs. The combined DME and CAFR configuration achieves higher AP than either individual modification in the reported ablation.

The proposed detector is deployed on an RK3566 edge device and integrated with machine monitoring and wired stop control. Defect inference and the stop decision execute locally without relying on cloud communication. Detection results are displayed locally for operators, while the cloud platform supports equipment monitoring and production quality analysis. The reported results establish test-set detection performance and describe the deployment workflow. Quantized-model accuracy, end-to-end response time, physical stopping delay, and false-stop frequency remain to be quantified before drawing conclusions about production performance. Future work will also evaluate generalization across knitted fabrics, texture patterns, and production conditions.

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